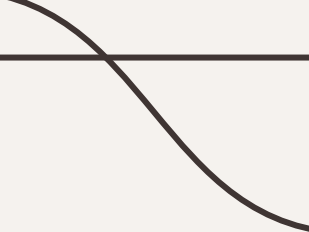


Predicting Song Popularity through Lyrics and Features

Sylvey Lin, Zhi-Yi Cao



Outline

01

Introduction

02

Dataset

03

Preprocessing

04

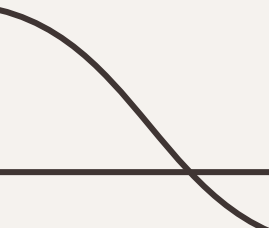
Models

05

Experiment

06

Conclusion





01

Introduction

Introduction

- Context
 - The music industry is growing rapidly.
 - Success requires understanding key factors influencing popularity.
 - Challenges
 - Personal experience can't match data-driven insights in predicting song popularity.
 - Objectives
 - How text mining can analyze song lyrics and features to predict popularity.
 - Dataset
 - Lyrics, titles, features, and artist data from the Genius Song Lyrics dataset.
 - Target Variable: Song Views
-

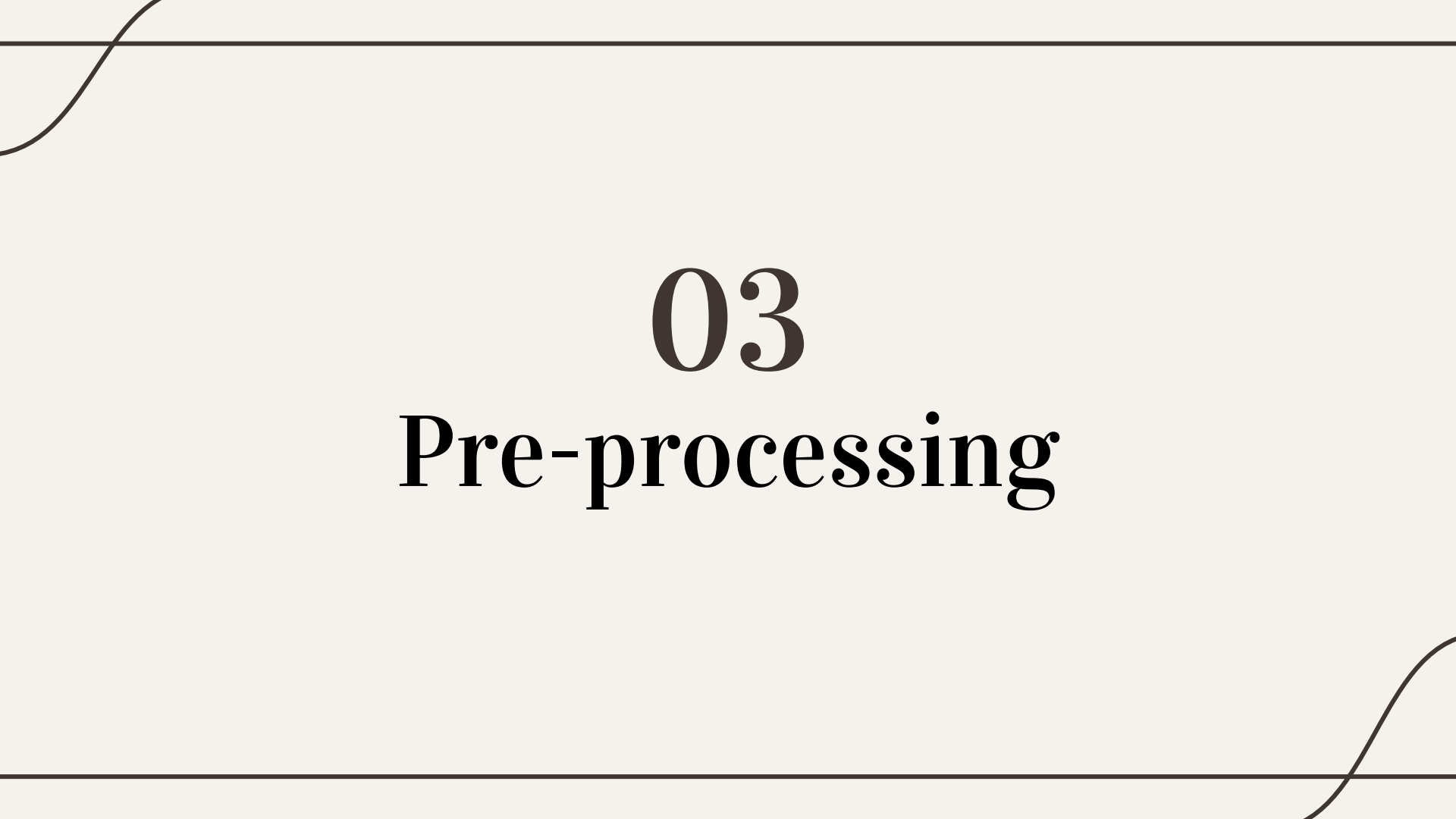


02

Dataset

Dataset

- Source of Dataset
 - Genius Song Lyrics dataset, sourced from Kaggle
 - Attributes
 - Title, **Tag**, year, Lyrics, Title, **Features**, Artist, Views, Id, language
 - Attributes used in the Project:
 - Input: Lyrics, Title, Features, Artist, Tags
 - Output: view counts
 - Data Characteristics
 - 3.37 million data points
 - Focused solely on songs in english
-



03

Pre-processing

Input

- Purpose of Preprocessing
 - Ensure that input features are clean, consistent for analysis.
 - Remove irrelevant elements to focus on meaningful data.
 - Steps in Input Preprocessing
 - Song Part Indicator Removal, ex: "[Intro]", "[Chorus]", "[Verse]" in lyric
 - Punctuation and Symbol Cleaning
 - Text Normalization
-

Input

- Original Data Example:

Title	Artist	Lyrics	Features
Killa Cam	Cam'ron	[Chorus: Opera Steve & Cam'ron] Killa Cam...	{"Cam'ron", "Opera Steve"}
Can I Live	JAY-Z	[Produced by Irv Gotti] [Intro] Yeah, hah...	{}
Forgive Me Father	Fabulous	Maybe cause I'm eatin And these bastards...	{}

- Cleaned Data Example:

Title	Artist	Lyrics	Features
Killa Cam	Cam'ron, Opera Steve	Killa Cam, Killa Cam, Cam...	[Cam'ron, Opera Steve]
Can I Live	JAY-Z	Yeah, hah, yeah, Roc-A-Fella...	[]
Forgive Me Father	Fabulous	Maybe cause I'm eating...	[]

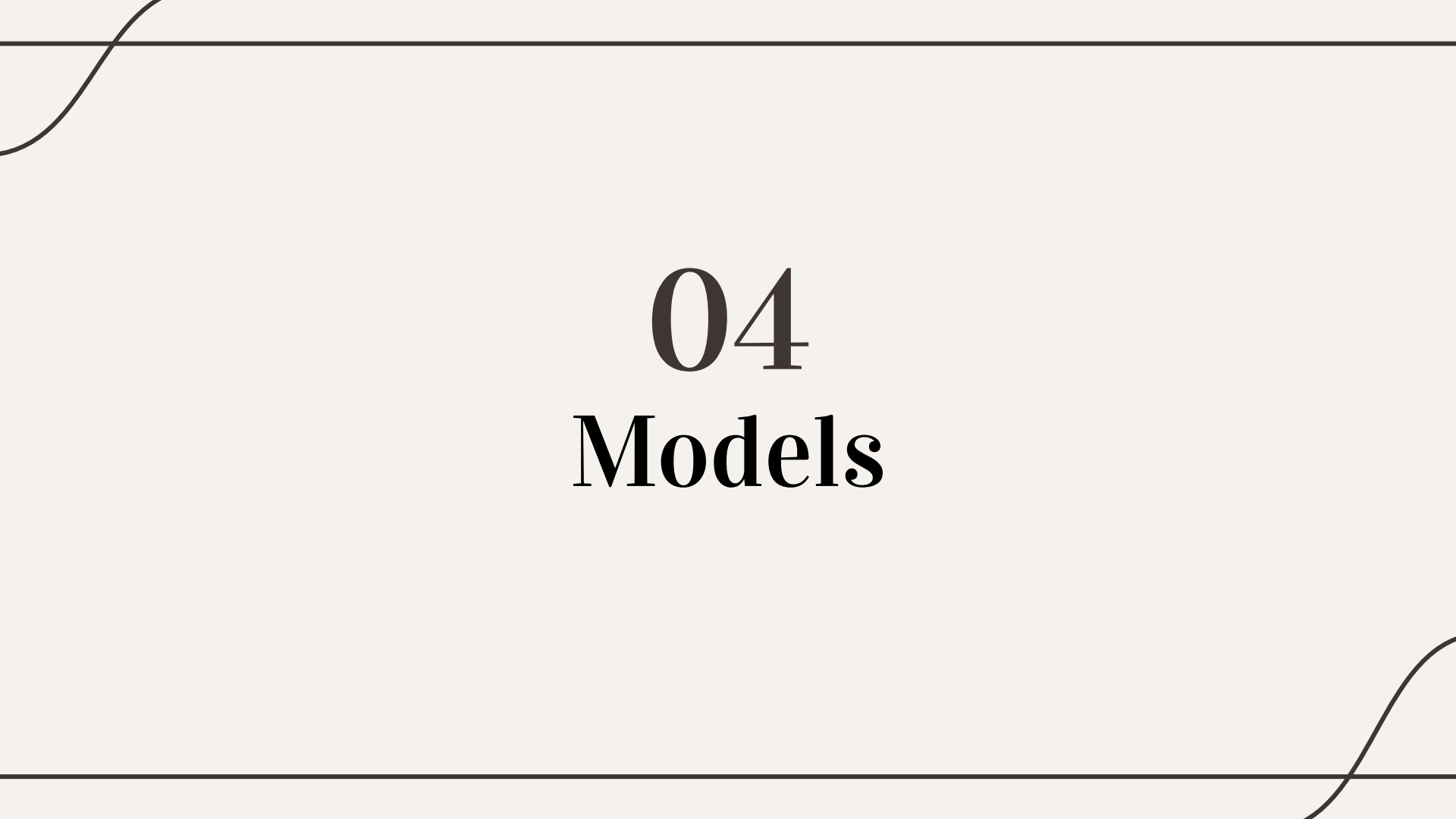
- Rationale: better models and reduce influence from noise

Output

Popularity_Bin

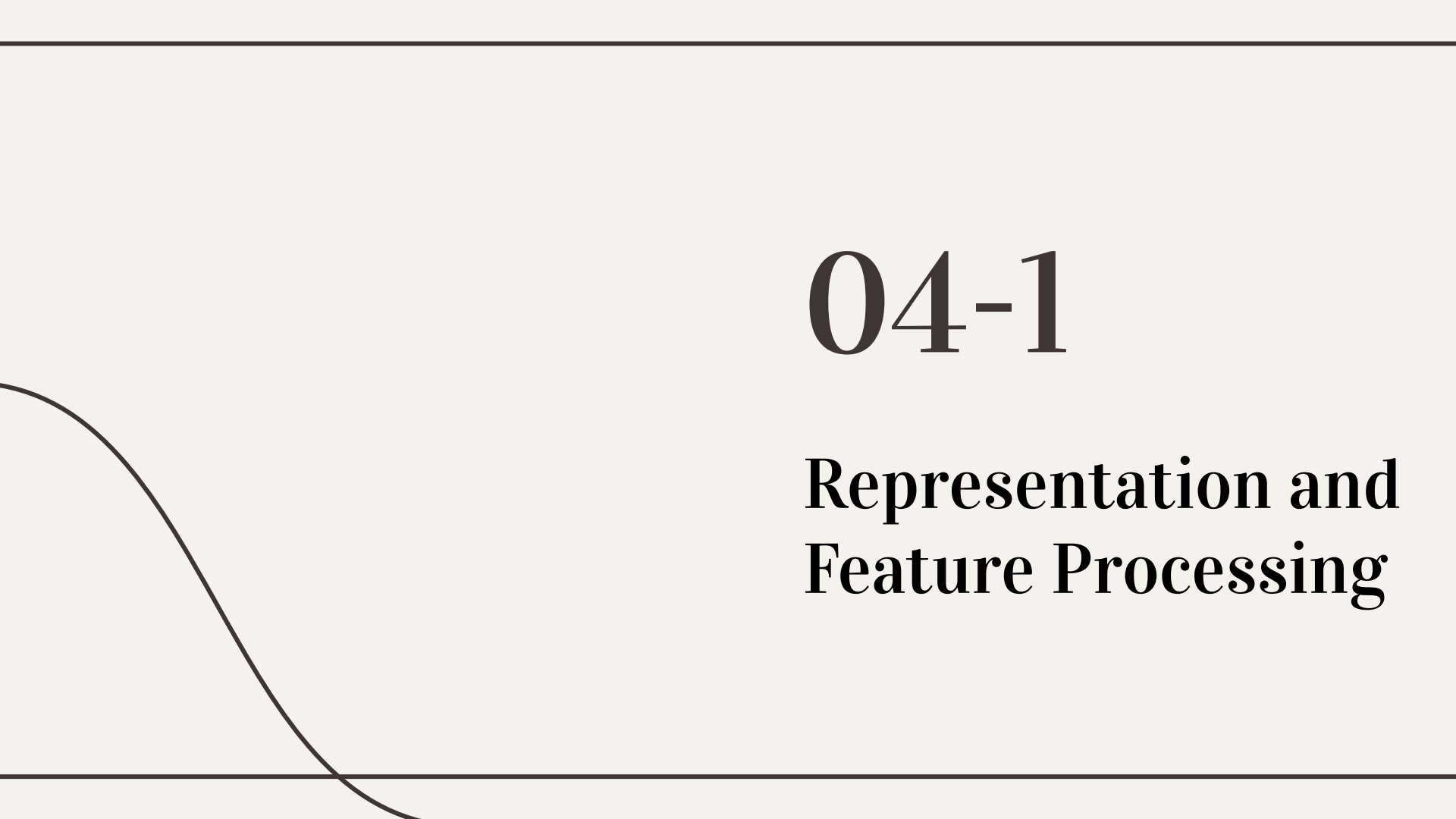
- Apply approximal equal-depth binning on “Views” of the whole dataset
 - `pandas.qcut()`
- Result count of each bin

Very Low	680,270
Low	672,183
Medium	675,187
High	672,114
Very High	674,444



04

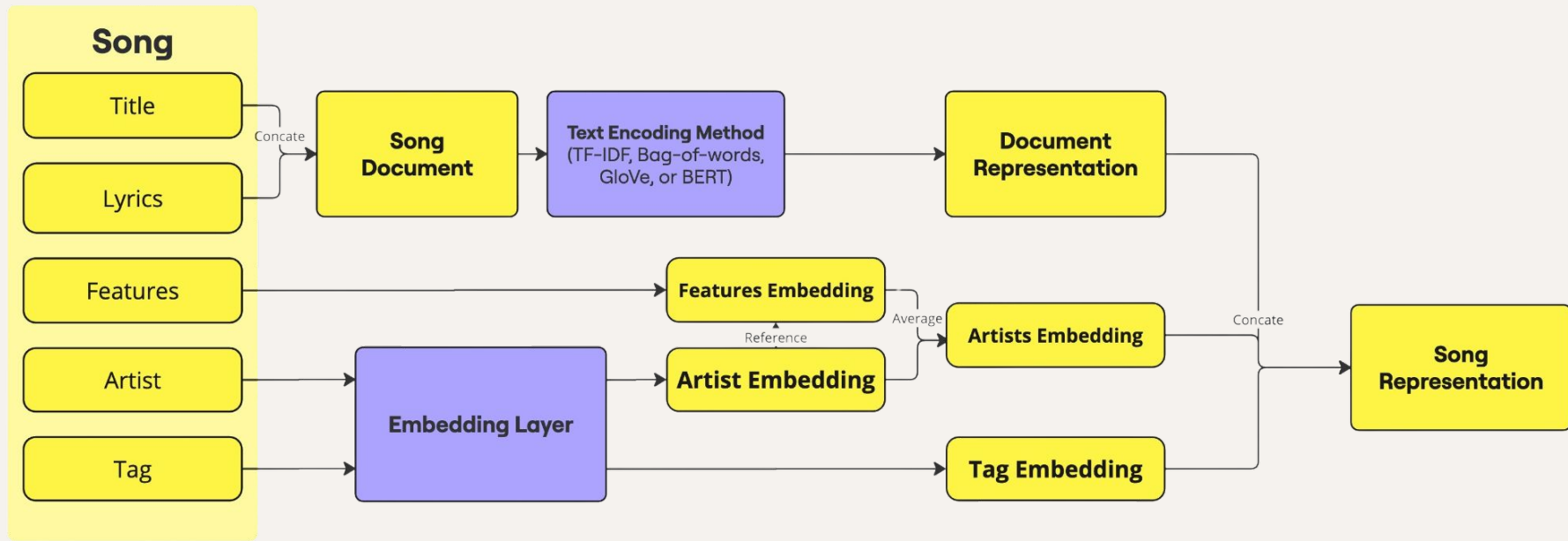
Models

A thin, dark horizontal line spans the top of the slide. On the left side, a dark, smooth curve starts from the top edge and sweeps downwards and to the right, ending near the bottom edge.

04-1

Representation and Feature Processing

Song representation generation process



Text Encoding Methods

- TF-IDF
- Bag of Words
- GloVe
 - ``en_core_web_md``
 - Output dimension: 300
- BERT
 - ``bert-base-uncased``
 - Fine-tuned with 100,000 song documents in our dataset
 - Output dimension: 768

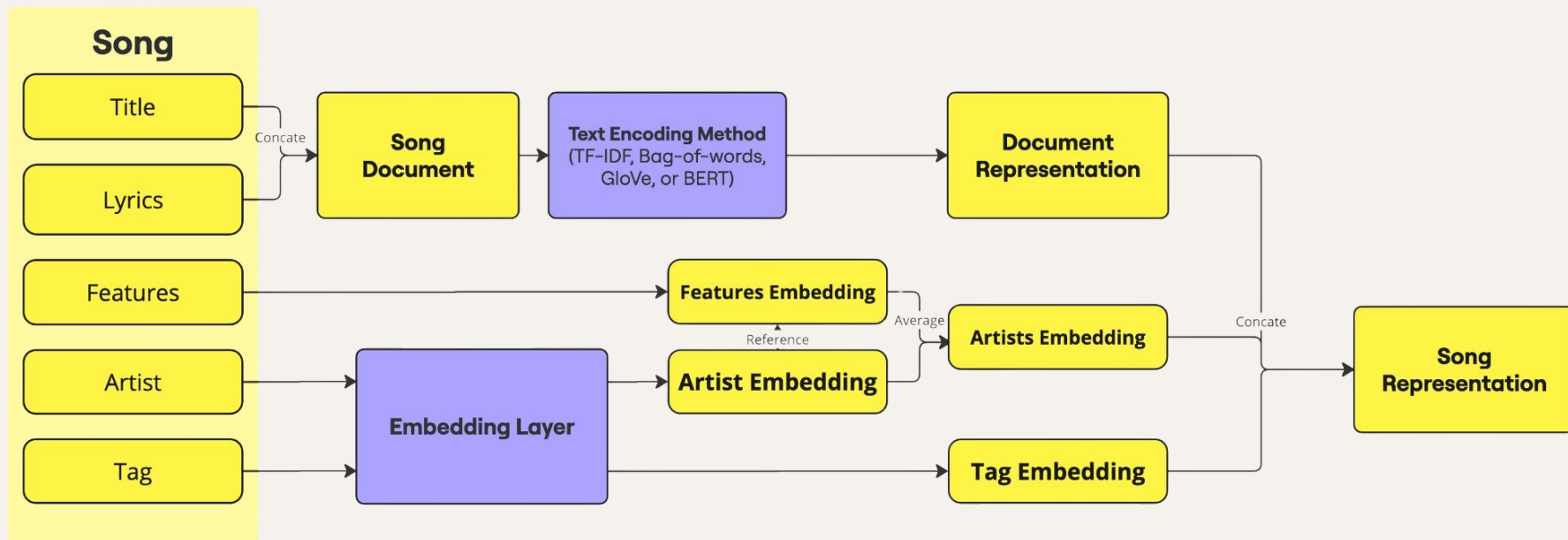
Tag Embedding

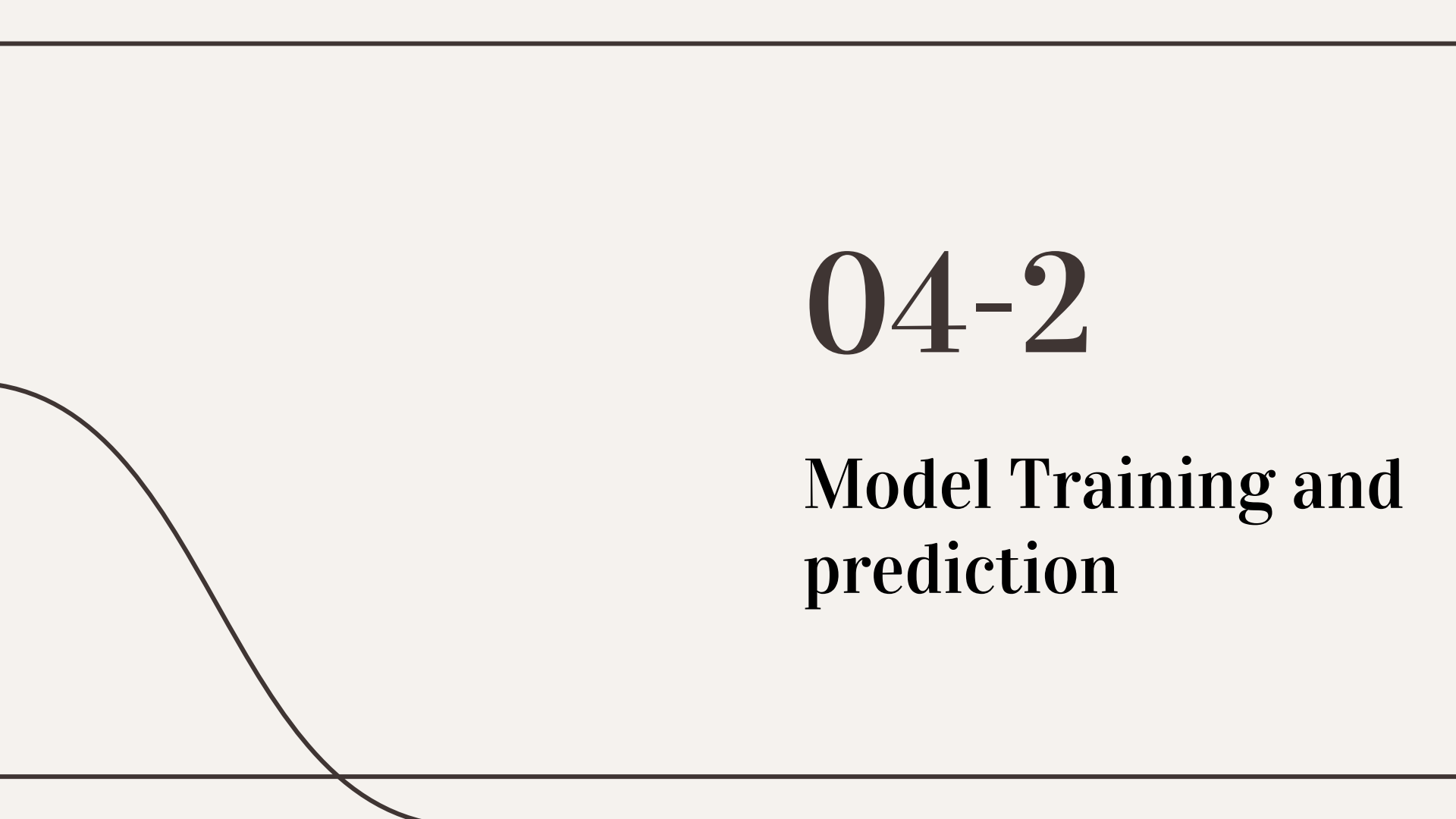
- Applied functions:
 - ``sklearn.preprocessing.LabelEncoder``
 - ``tf.keras.layers.Embedding``
- Output dimension: 2

Artists Embedding

- Artist Embedding
 - Applied functions
 - ``sklearn.preprocessing.LabelEncoder``
 - ``tf.keras.layers.Embedding``
 - Output dimension: 600
 - Feature Embedding
 - Reference Artist Embedding
 - Average Embedding of all featured artists embedding
 - Take the average of artist embedding and feature embedding to be Artists Embedding
-

Song representation generation process



A thin, dark horizontal line spans the top of the slide. On the left side, a dark, smooth curve starts from the edge and arcs downwards and to the right, ending near the bottom center of the slide.

04-2

Model Training and prediction

Models

- SVM
 - TF-IDF, Bag-of-words
 - Apply ``sklearn.svm.LinearSVC``
- Neural Network
 - GloVe, BERT
 - Popularity(y) encoding: Apply one-hot encoding
 - Number of layer: 2



05

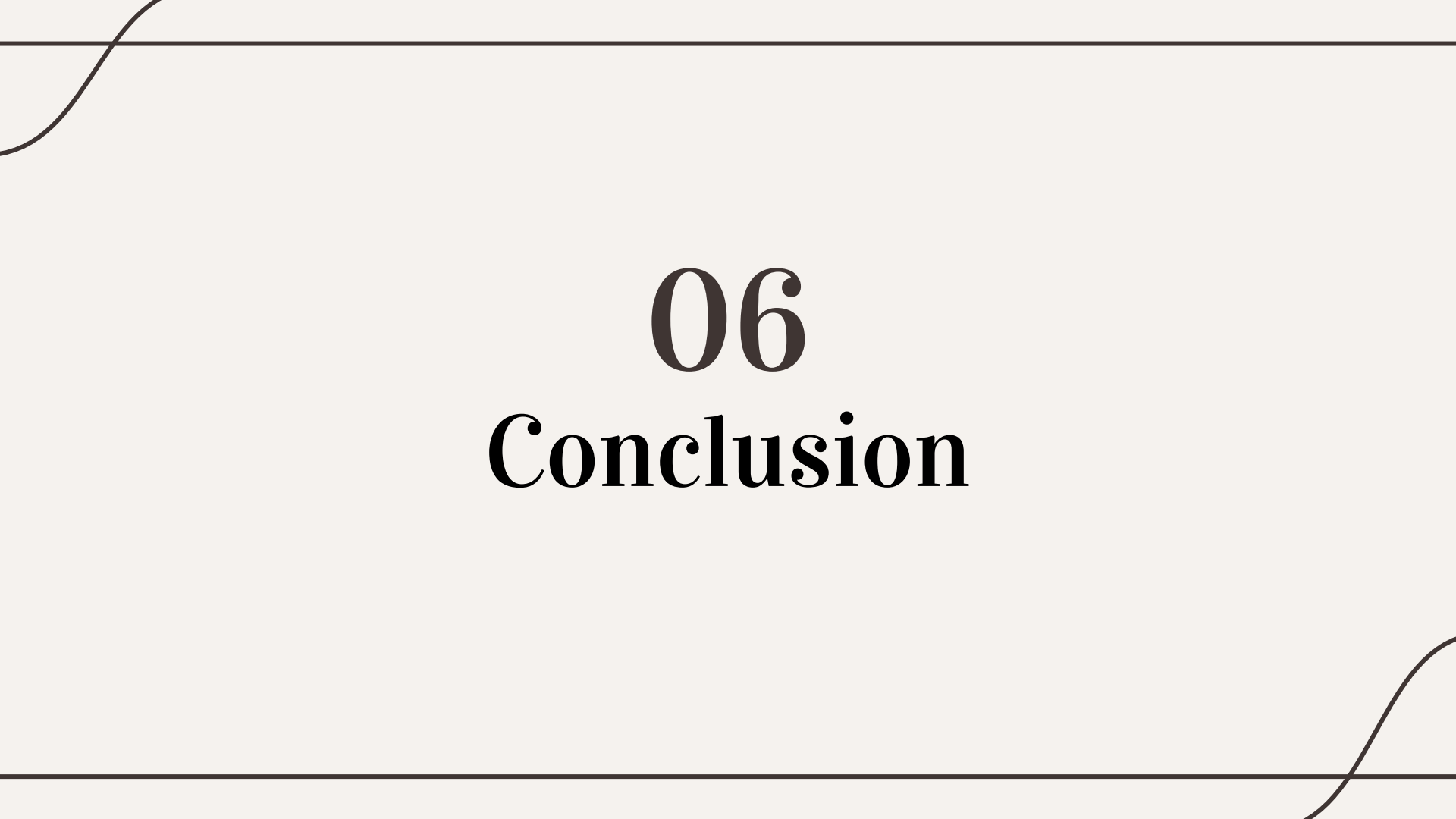
Experiment

Setup

- Data points:
 - 20,000 data points randomly selected per popularity bin (100,000 in total)
 - Training and testing ratio: 9:1
 - Neural Network setup
 - Optimizer: Adam
 - Loss: categorical cross entropy
 - Epochs: 20
 - Batch size: 32
 - Validation split: 0.2
-

Result

	Tag and artist	Accuracy	F1-score
TF-IDF + SVM		0.26	0.26
BoW + SVM		0.24	0.24
TF-IDF + SVM	✓	0.27	0.27
BoW + SVM	✓	0.25	0.25
GloVe + NN	✓	0.30	0.29
BERT + NN	✓	0.28	0.26



06

Conclusion

Conclusion

- Performance
 - Model performance was below expectations, with limited accuracy and F1-scores across experiments.
 - GloVe + Neural Network achieved slightly better results.

Future improvements

- Simplify Binning
 - Reduce the number of bins for more robust classification.
- Enhance Computational Resources
 - Utilize cloud-based platforms with Larger Ram and more powerful GPUs.
- Explore Advanced Methods
 - Experiment with ensemble models or fine-tuned transformers.
 - Integrate additional features like user demographics or temporal trends.